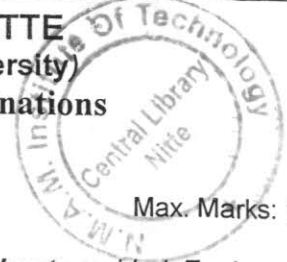


NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023
BT1002-1 - CELL BIOLOGY



Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- _____ the field where living systems are analyzed within the context of other systems.
 A) Biological System B) Bioinformatics
 C) Comparative Biology D) Systems Biology
- Which of the following is not a function of the cell membrane?
 A) Transport of substances into and out of the cell B) Protection of the cell
 C) Cell-to-cell communication D) Production of energy
- Which of the following organelles is responsible for the transport of proteins and lipids throughout the cell?
 A) Nucleus B) Mitochondria
 C) Endoplasmic reticulum D) Ribosomes
- Which organelles are found within the cytoplasmic matrix?
 A) Mitochondria and ribosomes B) Nucleus and Golgi apparatus
 C) Endoplasmic reticulum and lysosomes D) Vacuoles and chloroplasts
- The Golgi apparatus is named after the Italian scientist Camillo Golgi who discovered it in
 A) 1837 B) 1898
 C) 1920 D) 1954
- What is the difference between rough endoplasmic reticulum and smooth endoplasmic reticulum?
 A) Rough ER has ribosomes on its surface, while smooth ER does not
 B) Rough ER has protein on its surface, while smooth ER does not
 C) Rough ER has RNA on its surface while smooth ER does not
 D) Rough ER has granules on its surface, while smooth ER does not
- During which phase of the cell cycle does DNA replication occur?
 A) G1 B) S
 C) G2 D) M
- How many chromosomes do daughter cells have at the end of meiosis?
 A) The same number as the parent cell B) Twice the number of the parent cell
 C) Half the number of the parent cell D) None of the above
- The cytoskeleton provides which of the following functions in cells?
 A) Cell division B) Cell movement
 C) Maintaining cell shape D) All of the these
- Which of the following is not a type of cell-cell adhesion molecule?
 A) Cadherins B) Integrins
 C) Tubulins D) Selectins
- Which of the following is not a component of a signaling pathway?
 A) Ligand B) Receptor
 C) Effector D) Mitochondria
- Which type of signaling molecule is synthesized and released by neurons to act on a postsynaptic cell?
 A) Hormones B) Neurotransmitters
 C) Growth factors D) Cytokines
- What is the structure of a tyrosine kinase receptor?
 A) It has a single transmembrane domain and an intracellular catalytic domain.
 B) It has seven transmembrane domains and an extracellular ligand-binding domain.
 C) It has a single transmembrane domain and an extracellular ligand-binding domain.
 D) It has two transmembrane domains and an intracellular catalytic domain.

14. Which of the following is the primary role of microbes in fermentation?
 A) To convert glucose into carbon dioxide and water. B) To break down glucose into smaller molecules.
 C) To produce ATP through oxidative phosphorylation. D) To convert glucose into ethanol or lactic acid.
15. Which of the following is an example of a microbial indicator of soil quality?
 A) pH B) Soil moisture
 C) All of the these D) Organic matter content
16. In which year reverse transcriptase in retrovirus was first discovered
 A) 1970 B) 1967
 C) 1957 D) 1980
17. Who discovered the three major components of single nucleotide
 A) Francoise Raspail B) Phoebus Levene
 C) Frederick Sanger D) Kary Mullis
18. _____ is a measure of how much larger a microscope causes an object to appear.
 A) Focal length B) Objective
 C) Condenser D) Magnification
19. cellular extensions, extend and contract through the reversible assembly and contraction of actin subunits into microfilaments is called _____
 A) Pseudopodia B) Phagocytosis
 C) Cytoplasmic streaming D) Contraction
20. Which of the following is NOT a cell adhesion molecule
 A) Cadherin B) Keratin
 C) Integrin D) Selectin

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give an account on cell theory.	4	L1	1	1,12
	b) What is plasma membrane? Describe fluid mosaic model of plasma membrane.	6	L1	1	1,12
	c) Differentiate eukaryotic cell with a prokaryotic cell.	6	L1	1	1,12
2.	a) With a suitable diagram explain in detail the structure of nucleus. Add a note on the function of nucleus.	8	L1	1	1,12
	b) Illustrate briefly about Scanning Electron Microscopy technique.	8	L2	2	4,12
3.	a) Briefly describe about the cell cycle regulation and add a note on the checkpoints of cell cycle.	8	L2	2	4,12
	b) Describe prophase-I of meiotic cell division. Add a note on the significance of meiotic cell division.	8	L2	2	4,12
Unit – II					
4.	a) Write an explanatory note on microtubules.	8	L2	3	1,12
	b) Write a note on kinesins as a motor protein.	8	L2	3	1,12
5.	a) Illustrate the general principles of cell communication.	8	L2	4	1,12
	b) Write a note on G-Protein coupled receptors.	8	L2	4	1,12
6.	a) Brief out the different types of cell junctions.	8	L2	3	1,12
	b) Elaborate on voltage-gated ion channel.	8	L2	3	1,12
Unit – III					
7.	Illustrate on the applications of cells in:				
	a) Fermentation process	8	L2	5	1,12
	b) Medical diagnosis	8	L2	5	1,12
8.	a) What are stem cells? Explain the different types of stem cells. Add a note on its applications.	8	L2	5	1,12
	b) Give a brief note on applications, advantages and disadvantages of tissue engineering.	8	L2	5	1,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
